Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical –XI

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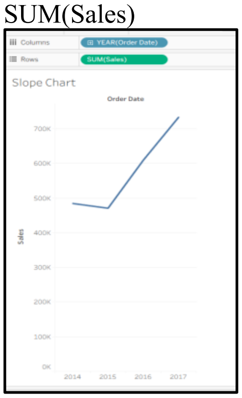
Class: TYIT	Batch: 1
Date of Assignment: 1/09/2026	Date/Time of Submission: 1/09/2026

Aim: Implementing Advanced Visualization Techniques and Customizations

A) Create slope charts and lollipop charts

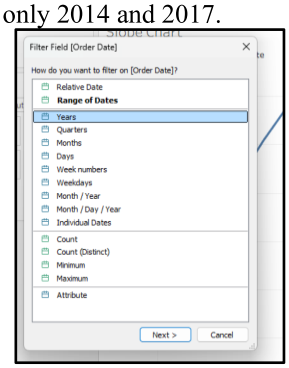
- **Create a Slope Chart (Year-over-Year Progression)(Set Up the Base Worksheet)**

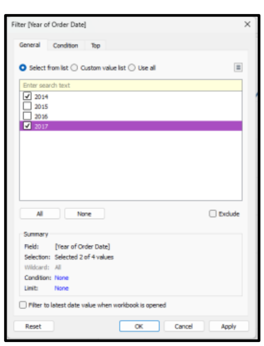
1. Open Tableau Desktop connected to Superstore.csv.
2. Create a new worksheet and rename it: Slope Chart.
3. Drag Order Date to the Columns shelf. (It will appear as YEAR(Order Date)).
4. Drag Sales to the Rows shelf (appears as SUM(Sales)



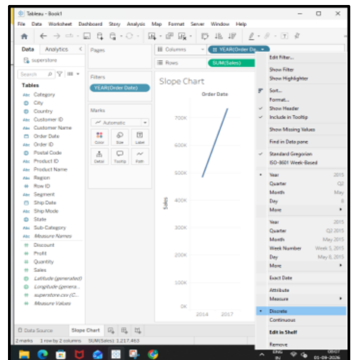
- Filter to Baseline & Comparison Years:

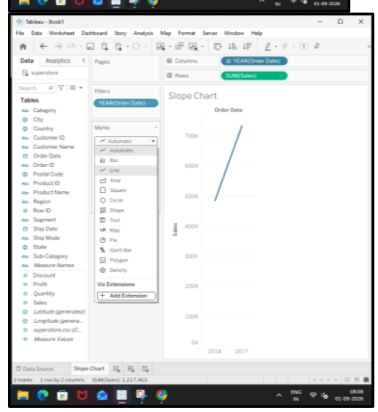
1.Drag Order Date to Filters, select Years, and keep only 2014 and 2017.





3.Right-click YEAR(Order Date) on Columns → select Discrete.

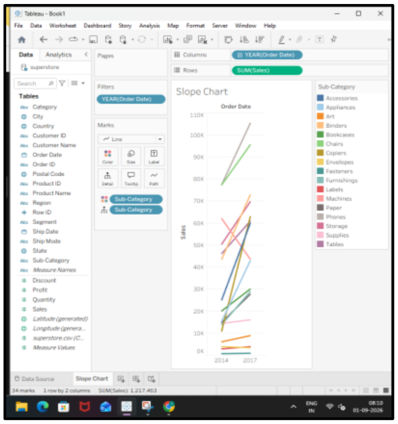




- Disaggregate by Sub-Category

1. On the Marks card, change the mark type dropdown from Automatic to Line.

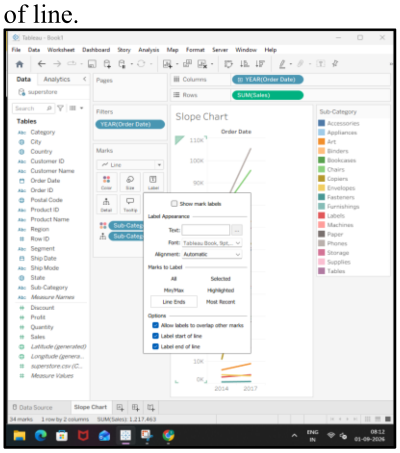
2. Drag Sub-Category onto the Detail tile (and onto the Color tile) on the Marks card.



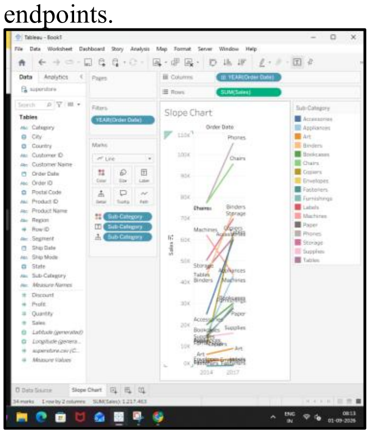
3. You will now see 17 distinct sloping lines connecting 2014 to 2017.

- Format Labels and Endpoints

1. On the Marks card, click the Label tile: o Check Show mark labels. o Under Marks to Label, choose Line Ends. o Check Label start of line and Label end of line.



2. Drag Sub-Category onto Label on the Marks card to display the product name directly at the line endpoints.

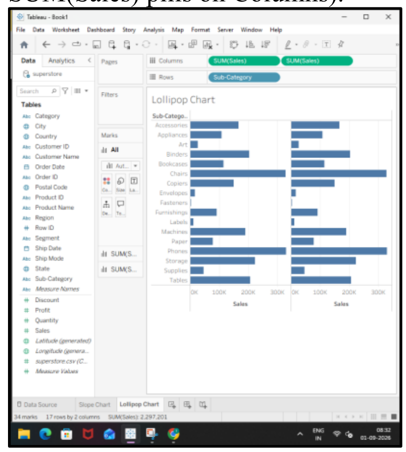


2. Create a Lollipop Chart (Sales with Profit Highlighting)

- Step 1: Set Up the Base Worksheet

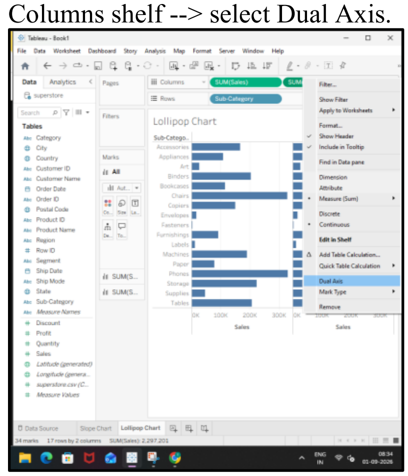
1. Create a new worksheet and rename it: Lollipop Chart. 2. Drag Sub-Category onto the Rows shelf.

3. Drag Sales onto the Columns shelf.
4. Drag Sales onto the Columns shelf a second time (creating two adjacent SUM(Sales) pills on Columns).



- Convert to Dual Axis

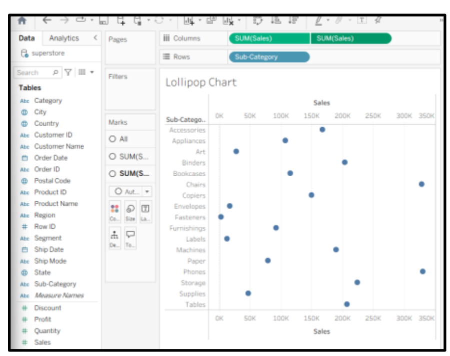
1. Right-click the second SUM(Sales) pill on the Columns shelf --> select Dual Axis.

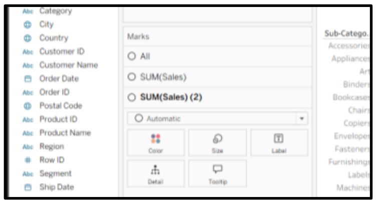


2. Right-click the top numerical axis in the visualization view --> select Synchronize Axis (crucial to ensure both axes align).

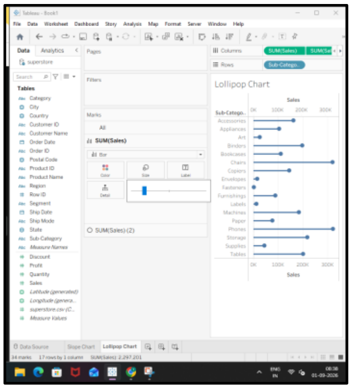
- Step 3: Configure the Marks Cards

In the Marks pane on the left, you will now see three cards: All, SUM(Sales) (first mark), and SUM(Sales) (2) (second mark).

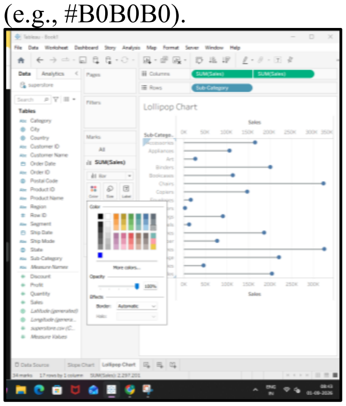




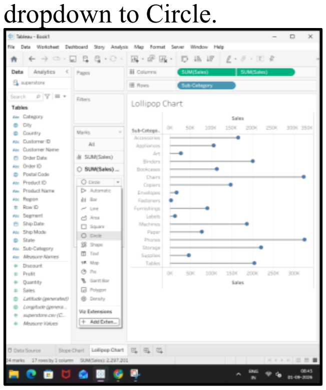
1. Configure the Stem (Bar): o Click on the first SUM(Sales) mark card. o Change the mark type dropdown to Bar. o Click the Size tile --> pull the slider down until the bars become thin, elegant stems.



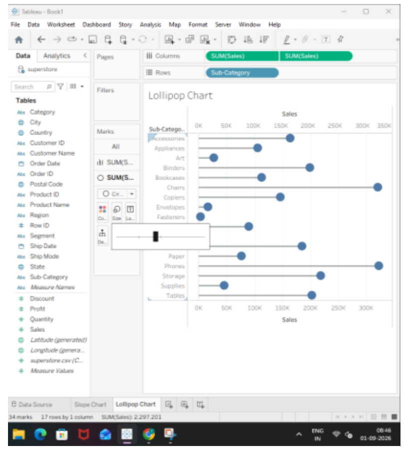
o Click Color --> set the bar color to a neutral gray (e.g., #B0B0B0).



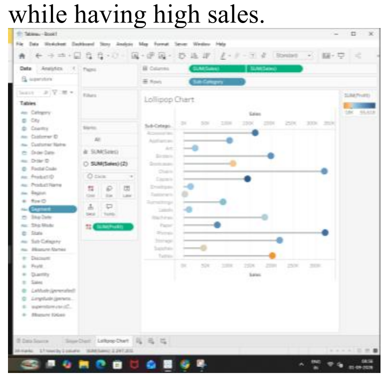
2. Configure the Head (Circle): o Click on the second SUM(Sales) (2) mark card. o Change the mark type dropdown to Circle.



o Click the Size tile --> increase the circle size so it sits neatly at the tip of each bar stem.

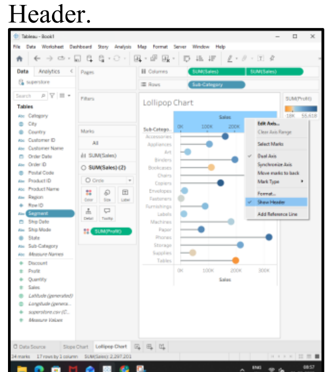


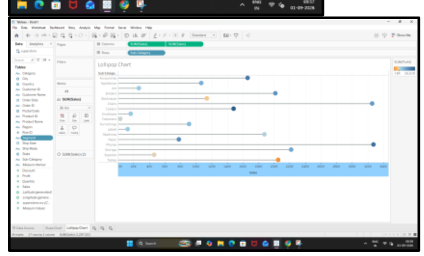
o Drag Profit onto the Color tile of this second marked card. o (Observation): Sub-categories like Tables immediately show negative profit (red/orange) while having high sales.



- Clean Up Axis Formatting

1. Right-click the top header axis --> uncheck Show Header.



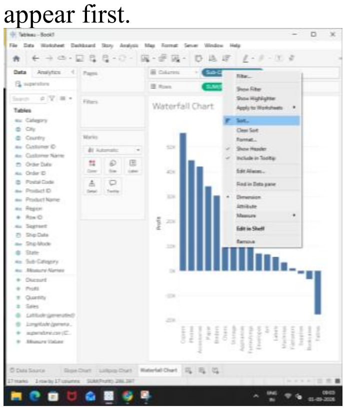


2. Click the Sort Descending button in the top toolbar to sort by total sales.

B) Design waterfall charts

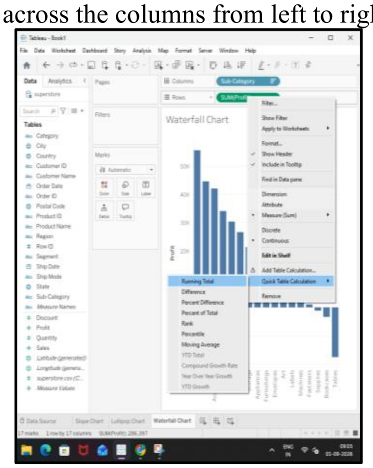
- **Set Up the Base Worksheet**

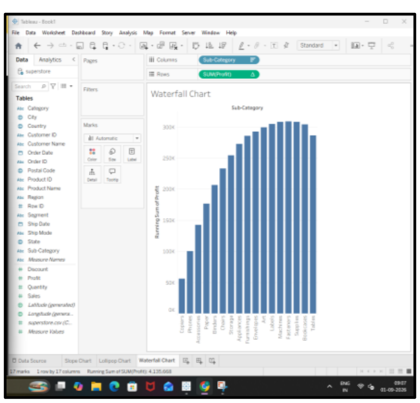
1. In Tableau Desktop, create a new worksheet and rename it: Waterfall Chart.
2. Drag Sub-Category onto the Columns shelf.
3. Drag Profit onto the Rows shelf (appears as SUM(Profit)).
4. Sort the view: Click the Sort Descending button on the top toolbar so the highest profit sub-categories appear first.



Apply Running Total Table Calculation

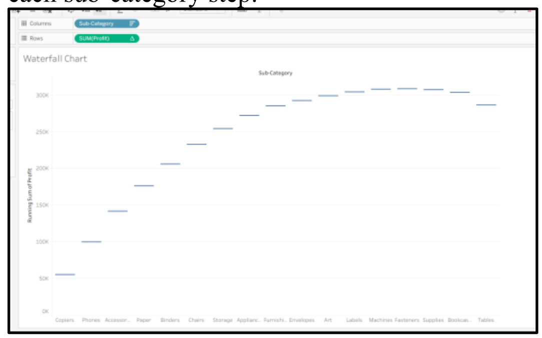
1. Right-click the SUM(Profit) pill on the Rows shelf.
2. Select Quick Table Calculation --> Running Total.
3. Notice that the line/bars now accumulate upward across the columns from left to right.





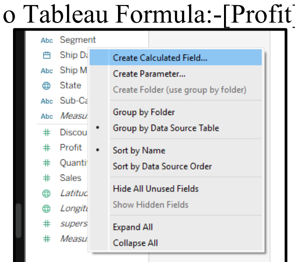
Switch Mark Type to Gantt Bar

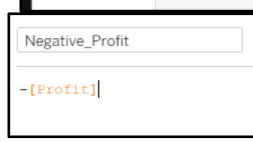
1. On the Marks card, change the dropdown selector from Automatic (or Bar) to Gantt Bar.
2. You will see thin horizontal baseline markers representing the running total cumulative level at each sub-category step.



Create the Negative Profit Sizing Calculation

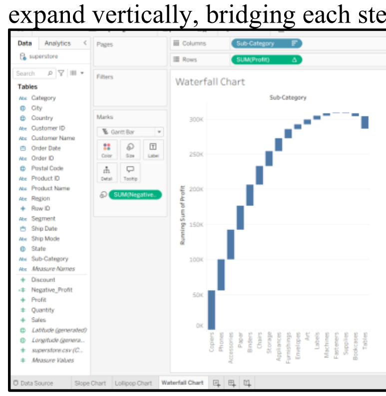
1. Set the field details: o Field Name: Negative_Profit , o Tableau Formula:-[Profit]





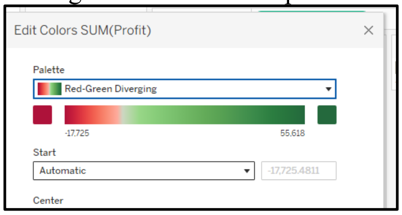
Size and Color the Waterfall Blocks

1. Drag Negative_Profit onto the Size tile on the Marks card. o The floating Gantt bars will now expand vertically, bridging each step to the next.



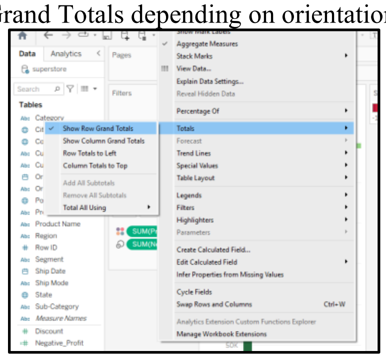
2. Drag Profit (original measure) onto the Color tile on the Marks card.
3. Click the Color tile --> select Edit Colors...: o Change Palette to Red-Green Diverging (or Red-Blue). o Check Use Full Color Range or set Center at 0. o Click OK. o (Observation: Profitable categories like Copiers/Phones appear green; loss-making

categories like Tables, Bookcases, and Supplies appear distinctly red,
showing where cumulative profit declines).

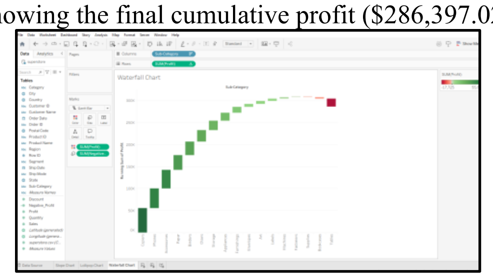


- Add Grand Totals to Close the Waterfall

1. Go to the top application menu: select Analysis --> Totals --> Show Row Grand Totals (or Show Column Grand Totals depending on orientation).



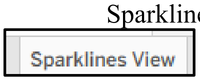
2. A final anchor pillar will appear at the right end showing the final cumulative profit ($286,397.02).



C) Build sparklines for trend analysis

Step 1: Create a new worksheet

1. Open Tableau Desktop.
2. Click New Worksheet at the bottom.
3. Rename the worksheet:
 Sparklines View



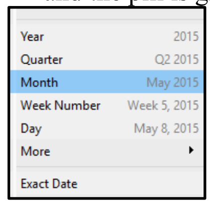
Step 2: Add Category and Sub-Category

From the Data Pane:

1. Drag Category → Rows.
2. Drag Sub-Category → Rows, to the right of Category.

Step 3: Add Order Date

1. Drag Order Date → Columns.
2. Right-click the Order Date pill.
3. Select the second Month option.
4. Make sure it is the continuous Month option and the pill is green.



Step 4: Add Sales

1. Drag Sales →Columns.
2. Tableau will create line charts for the different Sub-Categories.

You should now see the sparkline-style line charts.

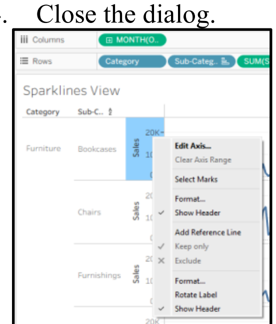
Step 5: Add Total Sales

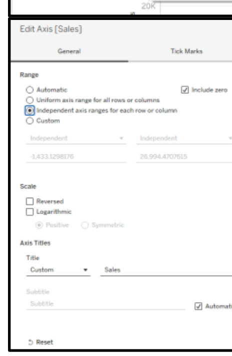
1. Drag Sales again.
2. Drop it on the Rows shelf, to the left of the existing SUM(Sales).

You can also use Sales on Tooltip/Label if you want a cleaner layout.

Step 6: Set Independent Axis

1. Right-click the Sales Y-axis.
2. Select Edit Axis...
3. Under Range, select:
 Independent axis ranges for each row or column
4. Close the dialog.





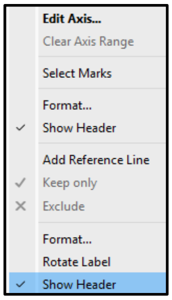
Step 7: Hide the Y-axis

1. Right-click the vertical Sales axis.

2. Uncheck Show Header.

Step 8: Hide the X-axis

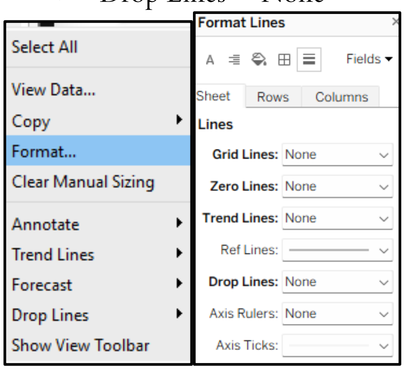
1. Right-click the bottom Order Date axis.
2. Uncheck Show Header.



Step 9: Remove Gridlines

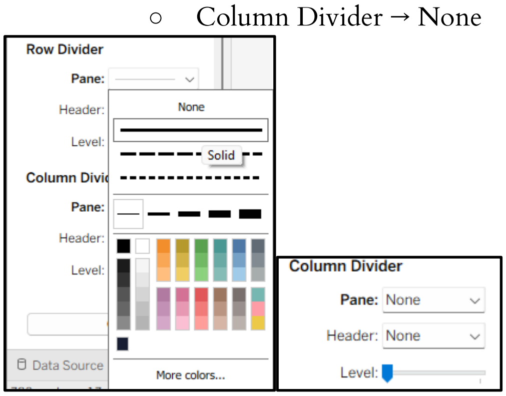
1. Right-click anywhere on the chart.
2. Select Format...
3. In the Format pane, click the Lines icon.
4. Set:

- Grid Lines → None
- Zero Lines → None
- Trend Lines → None
- Drop Lines → None



Step 10: Set Borders

1. Click the Borders icon in the Format pane.
2. Set:
   - Row Divider → Light thin line
   - Column Divider → None



Step 11: Show Minimum and Maximum Values

1. Go to the Marks card.
2. Click Color and set the line color to a dark color.
3. Click Label.
4. Check Show mark labels.
5. Under Marks to Label, select Min / Max.
6. Under Field, make sure SUM(Sales) is selected.
7. Check:
   - Label minimum value
   - Label maximum value

D) Implement sheet swapping for dynamic dashboards

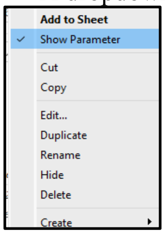
Step 1: Create Parameter

- Data Pane → Create Parameter
- Name: Select_View_Parameter
- Data Type: String
- Allowable Values: List
- Add:
  - Category Sales (Bar)
  - Regional Profit (Map)
- Current Value: Category Sales (Bar)
- Click OK.

Step 2: Show Parameter

- Right-click Select_View_Parameter
- Click Show Parameter.

- A dropdown appears.

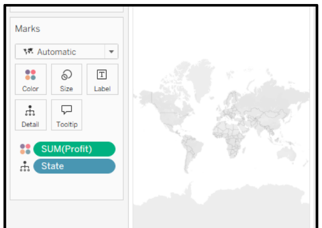


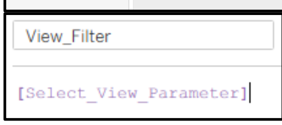
Step 3: Create Calculated Field

- Data Pane → Create Calculated Field
- Name: View_Filter
- Formula:

[Select_View_Parameter]

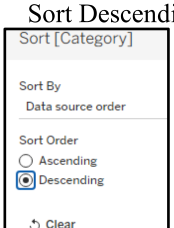
- Click OK.

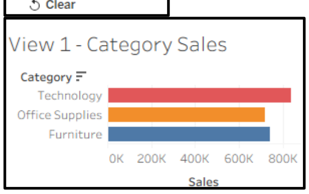




Step 4: Create View 1

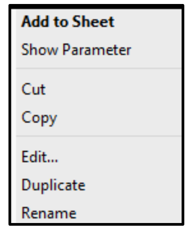
- New Worksheet → View 1 – Category Sales
- Category → Rows
- Sales → Columns
- Category → Color
- Sort Descending.

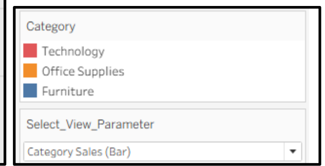


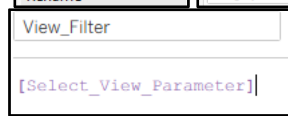


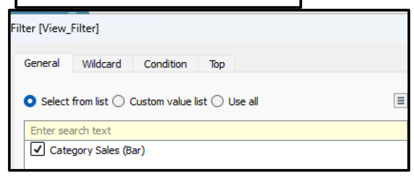
Step 5: Apply Filter

- Select parameter: Category Sales (Bar)
- View_Filter → Filters
- Select Category Sales (Bar) → OK.







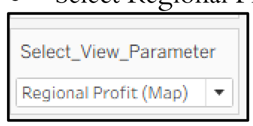


Step 6: Create View 2

- New Worksheet → View 2 – Regional Profit
- State → Canvas
- Profit → Color
- This creates the map.

Step 7: Apply Filter

- Select parameter: Regional Profit (Map)
- View_Filter → Filters
- Select Regional Profit (Map) → OK.



Step 8: Create Dashboard

- New Dashboard
- Drag Vertical Layout Container onto dashboard.

Step 9: Add Sheets

Inside the same container:

- View 1 - Category Sales
- View 2 - Regional Profit

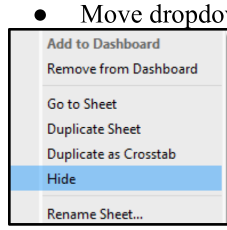
Step 10: Hide Titles

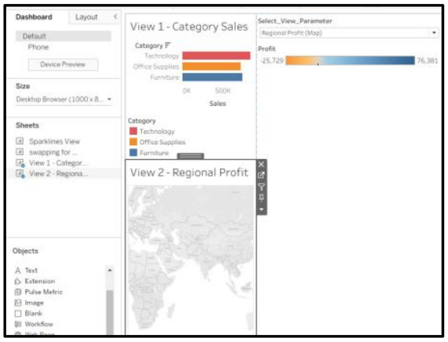
For both sheets:

Right-click Title → Hide Title.

Step 11: Add Dropdown

- Sheet dropdown arrow → Parameters
- Select Select_View_Parameter
- Move dropdown to top.



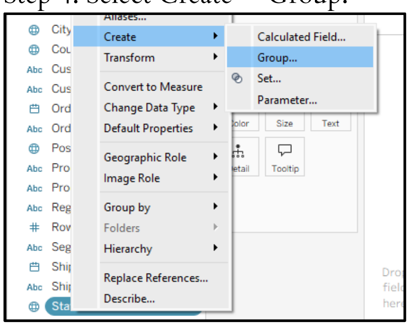


Step 12: Test

- Category Sales (Bar) → Bar chart appears.
- Regional Profit (Map) → Map appears.
- The other sheet collapses.

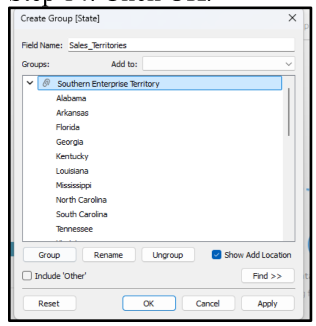
Aim : E) Create Custom Geographic Territories

Step 1: Create a Custom State Grouping (Territories)
Step 1: Open Tableau Desktop and connect the Superstore.csv dataset.
Step 2: Create a new worksheet and rename it Custom Territories Map.
Step 3: In the Data Pane, right-click State.
Step 4: Select Create → Group.

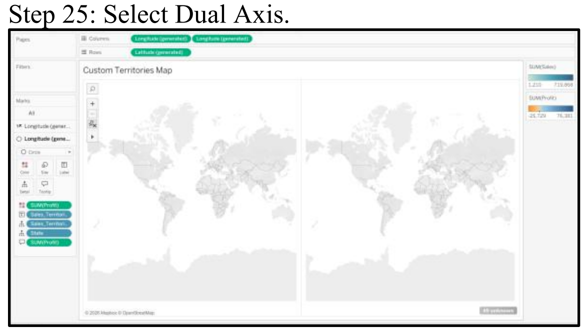


Step 5: Select these states and click Group:
California, Washington, Oregon, Nevada, Arizona, Utah, Idaho, Montana, Wyoming, Colorado and New Mexico.
Step 6: Rename this group Pacific & Mountain Zone.
Step 7: Select these states and click Group:
Texas, Illinois, Wisconsin, Nebraska, Michigan, Ohio, Indiana, Minnesota, Iowa, Missouri, Kansas, Oklahoma, North Dakota and South Dakota.
Step 8: Rename this group Midwest & Central Hub.
Step 9: Select these states and click Group:
New York, Pennsylvania, Massachusetts, New Jersey, Delaware, Maryland, Connecticut, Rhode Island, Vermont, New Hampshire and Maine.
Step 10: Rename this group Atlantic & North-East.
Step 11: Select the remaining southern states and group them together.
Step 12: Rename this group Southern Enterprise Territory.

Step 13: Change the group field name to Sales_Territories.
Step 14: Click OK.



Create the Map
Step 15: Double-click Sales_Territories to create the map.
Step 16: Change the Marks type to Map.
Step 17: Drag Sales to Color.
Step 18: Drag Profit to Tooltip.
Step 19: Drag Sales_Territories to Label.
Create Dual Axis Map
Step 20: Hold Ctrl and drag Longitude (generated) to create a second map.
Step 21: On the second Marks card, drag State to Detail.
Step 22: Change the Mark type to Circle.
Step 23: Drag Profit to Color.
Step 24: Right-click the second Longitude (generated).
Step 25: Select Dual Axis.

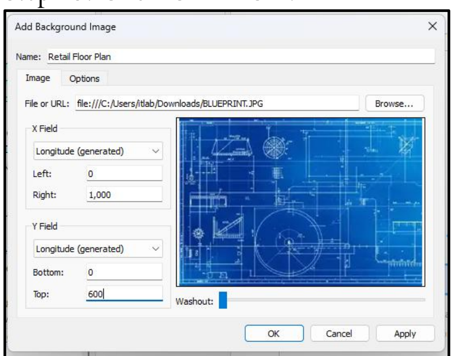


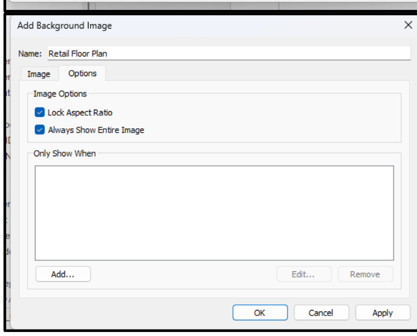
Aim: F) Use Background Images in Dashboards

Steps:

Step 1: Create a new worksheet.
Step 2: Rename it Store Floor Plan Heatmap.
Create X Coordinate
Step 3: Right-click in the Data Pane.
Step 4: Select Create Calculated Field.
Step 5: Enter the field name Store_X_Coordinate.
Step 6: Enter the X-coordinate formula given in the practical.
Step 7: Click OK.
Create Y Coordinate
Step 8: Create another Calculated Field.
Step 9: Enter the field name Store_Y_Coordinate.
Step 10: Enter the Y-coordinate formula given in the practical.

Step 11: Click OK.
Add Background Image
Step 12: Go to Map → Background Images → Superstore.
Step 13: Click Add Image.
Step 14: Enter the name Retail Floor Plan.
Step 15: Click Browse and select a floor plan or blueprint image.
Step 16: Select Store_X_Coordinate as the X Field.
Step 17: Set Left = 0.
Step 18: Set Right = 1000.
Step 19: Select Store_Y_Coordinate as the Y Field.
Step 20: Set Bottom = 0.
Step 21: Set Top = 600.
Step 22: Open the Options tab.
Step 23: Check Lock Aspect Ratio.
Step 24: Check Always Show Entire Image.
Step 25: Click OK → OK.

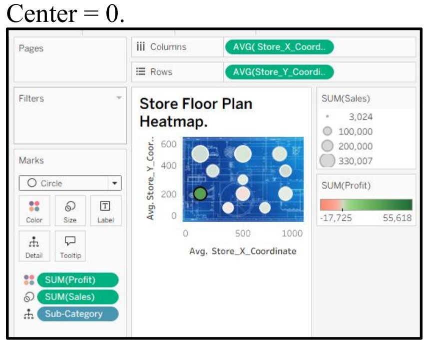




Create Floor Plan Visualization
Step 26: Drag Store_X_Coordinate to Columns.
Step 27: Drag Store_Y_Coordinate to Rows.
Step 28: The background image should appear.
Step 29: If the image does not appear, go to Analysis and uncheck Aggregate Measures.
Step 30: Right-click SUM(Store_X_Coordinate) and select Dimension or keep it as AVG.

Step 31: Right-click SUM(Store_Y_Coordinate) and select Dimension or keep it as AVG.

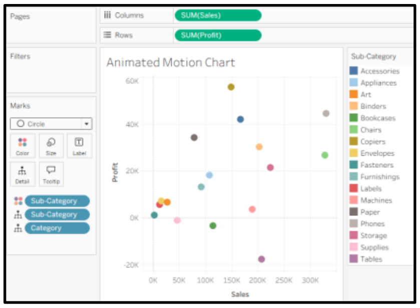
Add Sales and Profit
Step 32: Drag Sub-Category to Detail.
Step 33: Change the Mark type to Circle.
Step 34: Drag Sales to Size.
Step 35: Drag Profit to Color.
Step 36: Select Red-Green Diverging color with Center = 0.



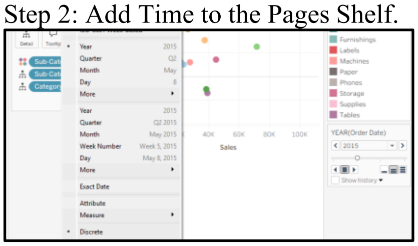
Add Labels and Clean the View
Step 37: Drag Sub-Category to Label.
Step 38: Check Show Mark Labels.
Step 39: Set label alignment to Bottom Center.
Step 40: Right-click the X-axis and uncheck Show Header.
Step 41: Right-click the Y-axis and uncheck Show Header.
Step 42: Set Grid Lines = None.

G: Apply Animation and Transparency Techniques

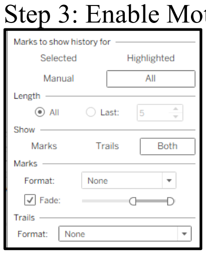
Task 1: Create an Animated Motion Chart (Pages Shelf)
Step 1: Set Up the Scatter Plot Worksheet



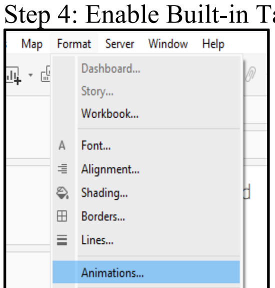
Step 2: Add Time to the Pages Shelf.

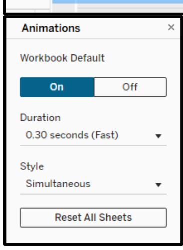


Step 3: Enable Motion History and Trails.

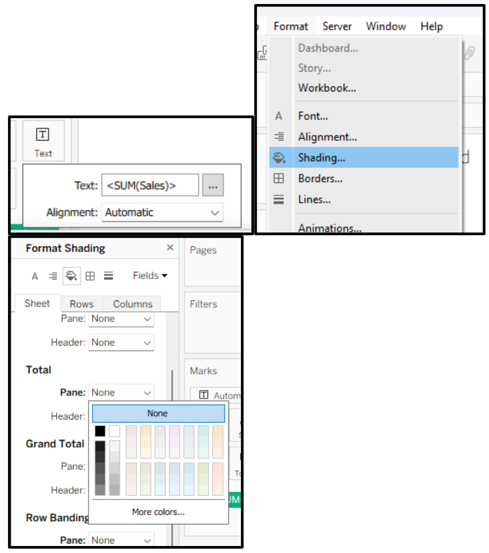


Step 4: Enable Built-in Tableau Client Animations.

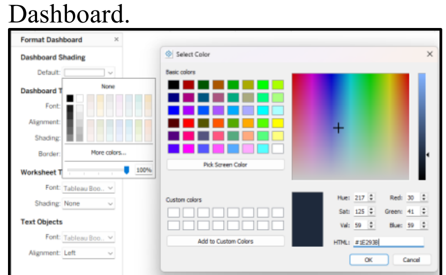


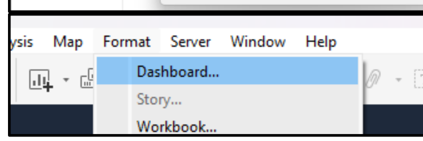


Task 2: Apply Sheet and Dashboard Transparency.
Step 1: Build a Transparent KPI Card Worksheet



Step 2: Assemble the Transparent Layered Dashboard.





Final Dashbord: